determine which path a signal photon takes to the screen by observing which down-converter spits out the idler-photon partner. And once again, the ability to glean which-path information about the signal photons—even though it is totally indirect, since we are not interacting with any signal photons at all—has the effect of preventing an interference pattern from forming.

Now for the weirder part. What if we manipulate the experiment so as to make it impossible to determine from which down-converter a given idler photon emerged? What if, that is, we erase the which-path information embodied by the idler photons? Well, something amazing happens: even though we've done nothing directly to the signal photons, by erasing the which-path information carried by their idler partners we can recover an interference pattern from the signal photons. Let me show you how this goes because it is truly remarkable.

Take a look at Figure 7.5b, which embodies all the essential ideas. But don't be intimidated. It's simpler than it appears, and we'll now go through it in manageable steps. The setup in Figure 7.5b differs from that of Figure 7.5a with regard to how we detect the idler photons after they've been emitted. In Figure 7.5a, we detected them straight out, and so we could immediately determine from which down-converter each was produced—that is, which path a given signal photon took. In the new experiment, each idler photon is sent through a maze, which compromises our ability to make such a determination. For example, imagine that an idler photon is emitted from the down-converter labeled "L." Rather than immediately entering a detector (as in Figure 7.5a), this photon is sent to a beam splitter (labeled "a"), and so has a 50 percent chance of heading onward along the path labeled "A," and a 50 percent chance of heading onward along the path labeled "B." Should it head along path A, it will enter a photon detector (labeled "1"), and its arrival will be duly recorded. But should the idler photon head along path B, it will be subject to yet further shenanigans. It will be directed to another beam splitter (labeled "c") and so will have a 50 percent chance of heading onward along path E to the detector labeled "2," and a 50 percent chance of heading onward along path F to the detector labeled "3." Now—stay with me, as there is a point to all this—the exact same reasoning, when applied to an idler photon emitted from the other down-converter, labeled "R," tells us that if the idler heads along path D it will